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Asymmetric contagion of jump risk in the Chinese financial sector: Monetary policy transmission matters

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ABSTRACT

Existing studies have identified the complexity of jump risk contagion but failed to make a clear distinction between upside and downside jump risks. This paper focuses on the asymmetric effects of jump risk contagion and explores its driving mechanisms from the perspective of monetary policy transmission. We decompose the bilateral jump risks in five financial subsectors into upside and downside jump risks using five-minute high-frequency data from December 13, 2018, to November 17, 2021 and capture their cross-sectional contagions in a network framework. The results indicate that downside jump risk is more contagious than upside jump risk. Furthermore, we verify that poor monetary policy transmission significantly enhances downside jump risk contagion but has no impact on upside jump risk contagion by constructing cointegration regressions. This work alerts risk managers to the complexity of jump risk contagion and suggests that policymakers prevent downside risk contagion by unblocking monetary policy transmission channels.

1. Introduction

Over the past several decades, financial markets have experienced tremendous perturbations that have led to extensive financial market shocks, such as the global financial crisis in the United States, the sovereign debt crisis in Europe, and the financial market turmoil brought about by the outbreak of the COVID-19 pandemic, among others, which have created a strong interest in researching the degree to which risk is interconnected. Most scholars have paid particular attention to problems that are "too-interconnected-to-fail" and have thus brought financial risk contagion to the fore (Allen and Gale, 2000; Elliott et al., 2014; Gofman, 2017).

A large body of financial literature has emerged on the dynamics of market risks. As the network analysis methods provide powerful tools for risk contagion research, most researchers have highlighted the role played by network connection in the evolution and interaction of financial risks. Billio et al. (2012) propose an econometric measure of connectedness based on a principal components analysis and Granger-causality directed risk networks; Diebold and Yılmaz (2014) construct directed and weighted network matrices using variance decomposition methods in the framework of vector autoregressive models; Silva et al. (2018) find that the network structure can either attenuate or amplify shocks from the real sector and thus plays a major role in risk contagion processes; Torri et al. (2021) develop a methodology based on conditional tail risk networks to assess the channels of risk transmission in the banking system. Other studies include, but are not limited to, Eisenberg and Noe (2001), Glasserman and Young (2015), Greenwood et al. (2015), Paltalidis et al. (2015), Poledna and Thurner (2016), Roy and Sinha Roy (2017), Kilic (2017), Greenwood-Nimmo et al. (2019), Buse and Schienle (2019), Wen et al. (2019), and Hu and Fan (2022). The related literature on risk contagion is still developing.

The main drawback of the above literature is that most researchers capture risk information from low-frequency data to conduct contagion analyses. However, it can be observed that unexpected intraday shocks in the stock market occur frequently. It is common for an unexpected jump in one stock market to cause another market to similarly jump. These sudden and extreme price change risks that occur in the stock market are defined as jump risks. Fortunately, some scholars have noticed correlation characteristics of jump risks such as Asgharian and Nossman (2011), who find that a large percentage of jumps in local markets are in response to US market and the regional market. Jawadi et al. (2015) examine the interaction of jumps between European markets and the US market. Similarly, Li et al. (2017) investigate the jump spillover effects between oil prices and exchange rates. Gong et al. (2020) analyze the contagion network of jump risks across different sectors in China.

These above studies on the contagion of jump risk generalize all jumps as being the same type of risky behavior and thus ignore the

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differences between upside and downside jump risks. In practice, risk refers to the uncertainty, including not only that of losses but also of gains, on financial asset returns. However, investors restricted from shorting in the Chinese stock market are generally averse only to the uncertainty of losses associated with downside movements of financial assets. Therefore, downside risk, particularly the extreme downside risk, is of concern to investors. Moreover, several collective financial market crashes in recent years have heightened the vigilance of Chinese financial market regulators regarding left-tail events.

There are also several studies that have found asymmetric effects of upside and downside risks. For example, Patton and Sheppard (2015) find that negative jumps in price lead to significantly higher future volatility. Baruník et al. (2016) also provide ample evidence of the asymmetric connectedness of "bad" and "good" volatility. Therefore, we argue that the contagion structures between upside and downside jump risks should, to a large extent, be distinct from each other. Moreover, it is vital to explore the driving mechanisms that cause asymmetric jump risk contagion. Although some existing literature has found the drivers of cross-sectoral risk contagion (e.g., economic policy uncertainty, the COVID-19 pandemic outbreak, market liquidity, industry characteristics, quantitative easing, etc.) (e.g.,Bernal et al., 2016; Guo et al., 2021; Iwanicz-Drozdowska et al., 2021; Lien et al., 2022; Cao, 2022), they have not been able to explain the asymmetry between different types of jump risk contagion. Our research is therefore devoted to addressing this challenge by uncovering the driving mechanisms behind asymmetric jump risk contagion.

Economists and risk managers should not blur the distinction between upside and downside risk contagion. This paper is concerned with the asymmetric contagion effects of jump risk and its formation. We first quantify various jump risks (including bilateral, upside, and downside jump risks) of five financial subsectors – namely, the banking, insurance, securities, diversified finance, and real estate sectors in China – and provide a full exploration of their cross-sectoral contagions following Diebold and Yılmaz (2014)'s directed and weighted network framework. The network visualization clearly demonstrates the complexity of different jump risk contagions as well as how each financial subsector plays a different role in the dynamic networks. Furthermore, the measured downside jump risk is found to be more contagious than the upside jump risk. Subsequently, we pay particular attention to the mechanisms that contribute to this asymmetric effect. Our explanation for this phenomenon is that the blocked monetary policy transmission causes funds to stay in the financial system rather than be injected into the real economy, where it can be used to support economic activity. Prolonged stagnation of funds in the financial system promotes speculative activity and the formation of bubbles. It should note that financial bubbles usually form slowly, but they also burst systematically. Therefore, jump risk contagion can exhibit asymmetry. Through regression analysis, we show that poor monetary policy transmission significantly promotes downside jump risk contagion. However, upside jump risk contagion is not sensitive to monetary policy transmission.

Overall, our work is one of the few that focuses on analyzing different types of financial jump risks within a network framework using five-minute high-frequency data from Chinese financial markets. Furthermore, we characterize the cross-sectoral contagions of various jump risks in China. As such, this work enriches the existing empirical literature on extreme risk networks. We further fill a gap in the literature by focusing on the role of monetary policy transmission in the contagion of jump risks and verifying that poor monetary policy transmission promotes downside jump risk contagion but has no impact on upside jump risk contagion. In practice, our work serves as a reference for risk managers and policymakers. We suggest that risk managers incorporate the dynamic contagion indices constructed in our network analysis into their risk warning systems to improve the accuracy of their risk management tools. As for macro policymakers, this study highlights the importance of the undesirable impact of weak monetary policy transmission on risk contagion. On the one hand, they should not implementing excessively loose quantitative monetary policies, which will result in an oversupply of monetary liquidity. On the other hand, they should unblock the monetary policy transmission channel as soon as possible to facilitate the injection of funds into the real economy and prevent them from stagnating in the financial system and encouraging excessive financial speculation and asset bubbles formation.

The rest of the paper is organized as follows. In Section 2, we introduce the methods of jump risk identification and network construction as well as the measurement of risk contagion. Section 3 discusses the empirical results. Section 4 presents the conclusions and policy implications.

2. Methodology

2.1. Jump risk identification

We assume that the logarithm of the closing stock price of financial asset follows a standard jump-diffusion process

$$dp_t = \mu_t dt + \sigma_t dW_t + J_t dq_t \tag{2.1}$$

where p_t is the logarithm of the asset price, μ_t is the instantaneous drift rate, σ_t is the instantaneous volatility, W_t is the Wiener process, J_t is a process determining the jump size, and q_t is a counting process whose differential determines the times of the jump occurrences.

The total variability of the stochastic process during the period $t-1$ to t can be expressed in terms of its *quadratic variation* as

$$QV_t := IV_t + QJ_t = \int_{t-1}^{t} \sigma_s^2 dt + \sum_{t-1<s\leq t} \kappa_s^2 \tag{2.2}$$

where $\kappa_t = J_t \cdot \mathbb{1}_{\{q_t=1\}}$, $\mathbb{1}_{\{\cdot\}}$ is the indicator function. The first term, IV_t, in Eq. (2.2) represents the impact of the continuous price variability, which is called *integrated variance*, while the second term, QJ_t, namely *jump variance*, represents the impact of the discontinuous price variability.

Following Andersen et al. (2003, 2007), Barndorff-Nielsen and Shephard (2004), we compute the *realized variation* (RV_t) and *realized bipower variation* (BV_t), as $n \to \infty$

$$RV_t := \sum_{i=1}^{n} \Delta p_{i,t}^2 \xrightarrow{p} QV_t \tag{2.3}$$

$$BV_t := \frac{\pi}{2} \sum_{i=2}^{n} |\Delta p_{i,t}||\Delta p_{i-1,t}| \xrightarrow{p} IV_t \tag{2.4}$$

where the intraday log-return over the ith interval is defined as $\Delta p_{i,t} := p_{i\Delta_n,t} - p_{(i-1)\Delta_n,t}$, the sampling interval is denoted by Δ_n, and the number of intraday observations is n. Then, the jump variance can be estimated consistently by

$$JV_t := RV_t - BV_t \tag{2.5}$$

where JV_t is called *realized jump variance*, and $JV_t \xrightarrow{p} QJ_t$ as $n \to \infty$.

Referring to Barndorff-Nielsen and Shephard (2006), we construct the test statistic under the null hypothesis of no jumps

$$BNS_t := \frac{\Delta_n^{-1/2}}{\sqrt{(\frac{\pi^4}{4} + \pi - 5) max\{1, \frac{QP_t}{(BV_t)^2}\}}} (1 - \frac{BV_t}{RV_t}) \xrightarrow{d} N(0,1) \tag{2.6}$$

where $QP_t := n\frac{4}{\pi^2}\frac{n}{n-3}\sum_{j=4}^{n} |\Delta p_{j-3,t}||\Delta p_{j-2,t}||\Delta p_{j-1,t}||\Delta p_{j,t}|$.

Then, we identify the *significant jump variance* per day on a given confidence level $\alpha\%$ as

$$SJV_t = JV_t \cdot \mathbb{1}_{\{BNS_t > \Phi_\alpha^{-1}\}} \tag{2.7}$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function, and Φ_α^{-1} is the quantile function of the standard normal distribution.

Last, we measure the bilateral jump risk by $BJ_t = sign(r_t) \times \sqrt{SJV_t}$, where r_t is daily return and $sign(\cdot)$ is the sign function. Moreover, we have the upside jump risk measure, PJ_t, defined as $BJ_t \times \mathbb{1}_{\{BJ_t>0\}}$, and the downside jump risk measure, NJ_t, defined as $BJ_t \times \mathbb{1}_{\{BJ_t<0\}}$, respectively.

Table 1
Connectedness table.

	x_1	x_2	–	x_N	**From others**
x_1	$\tilde{D}^H_{11}$	$\tilde{D}^H_{12}$	–	$\tilde{D}^H_{1N}$	$Cidx^H_{1\cdot}$
x_2	$\tilde{D}^H_{21}$	$\tilde{D}^H_{22}$	–	$\tilde{D}^H_{2N}$	$Cidx^H_{2\cdot}$
–	–	–	–	–	–
x_N	$\tilde{D}^H_{N1}$	$\tilde{D}^H_{N2}$	–	$\tilde{D}^H_{NN}$	$Cidx^H_{N\cdot}$
To others	$Cidx^H_{\cdot 1}$	$Cidx^H_{\cdot 2}$	–	$Cidx^H_{\cdot N}$	$\boldsymbol{Cidx^H}$
Net contribution	$Cidx^H_{\cdot 1} - Cidx^H_{1\cdot}$	$Cidx^H_{\cdot 2} - Cidx^H_{2\cdot}$	–	$Cidx^H_{\cdot N} - Cidx^H_{N\cdot}$	

Note: This table is referenced from Diebold and Yılmaz (2014). The upper-left N × N block shows the results of generalized variance decompositions. The rightmost column gives the total directional connectedness (i.e., the contribution from all others to *i*); The penultimate row gives the total directional connectedness (i.e., the contribution to all others from *j*); The bottommost row gives the difference in total directional connectedness (i.e., the net contribution to all others). The bottom-right element is the total connectedness.

2.2. Network construction and risk contagion measurement

Following Diebold and Yılmaz (2012, 2014), we consider a covariance stationary N-variable VAR(p) process as

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + \varepsilon_t \tag{2.8}$$

where $X_t = [x_{1t}, x_{2t}, \dots, x_{Nt}]'$, and $\varepsilon_t \sim N(0, \Sigma)$. Equivalent, it can be re-written as a $VMA(\infty)$ process

$$X_t = \sum_{i=1}^{\infty} A_i \varepsilon_{t-i} \tag{2.9}$$

where the N-dimensional coefficient matrices A_i follow the recursion $A_i = \phi_1 A_{i-1} + \phi_2 A_{i-2} + \dots + \phi_p A_{i-p}$ with A_0 being an identity matrix and $A_i = 0$ for $i < 0$.

Then, we use generalized variance decomposition (Pesaran and Shin, 1998) to assess the proportion of the H-step ahead forecast error variance of x_i which is accounted for by the innovations in x_j, and $\forall j \neq i$ for each i.

$$D^H_{ij} := \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_i)} \tag{2.10}$$

where σ_{jj} is the standard deviation of the error term for the jth equation, and e_i is the selection vector, with one as the ith element and zeros otherwise.

Moreover, we normalize the variance decomposition matrix by

$$\tilde{D}^H_{ij} := \frac{D^H_{ij}}{\sum_{j=1}^{N} D^H_{ij}} \times 100 \tag{2.11}$$

By construction, $\sum_{j=1}^{N} \tilde{D}^H_{ij} = 1$ and $\sum_{i=1}^{N} \sum_{j=1}^{N} \tilde{D}^H_{ij} = N$.

Also, we construct the total connectedness index and two directional connectedness indices as follows:

$$Cidx^H := \frac{\sum_{i,j=1; i\neq j}^{N} \tilde{D}^H_{ij}}{N} \tag{2.12}$$

$$Cidx^H_{i\cdot} := \sum_{j=1; i\neq j}^{N} \tilde{D}^H_{ij} \tag{2.13}$$

$$Cidx^H_{\cdot j} := \sum_{i=1; i\neq j}^{N} \tilde{D}^H_{ij} \tag{2.14}$$

Intuitively, in Table 1 called the connectedness table, the upper-left N×N block shows the results of generalized variance decompositions. The rightmost column gives the total directional connectedness (i.e., the contribution from all others to *i*) by Eq. (2.13); The penultimate row gives the total directional connectedness (i.e., the contribution to all others from *j*) by Eq. (2.14); The bottommost row gives the difference in total directional connectedness (i.e., the net contribution to all others). The bottom-right element is the total connectedness computed by Eq. (2.12). In subsequent regression analysis, we use $Cidx^H$ to measure the level of jump risk contagion throughout the financial sector.

3. Empirical analysis

3.1. Jump risk identification

Our sample data contain five financial sector indices in China, including the Securities Sector Index, the Banking Sector Index, the Insurance Sector Index, the Diversified Finance Sector Index, and the Real Estate Sector Index.[1] The sample time series is provided by the WIND financial database and ranges from December 13, 2018, to November 17, 2021, for a total of 709 trading days and consists of five-minute prices from 9:30 to 15:00, Beijing Time.

We first apply the methodology described in Section 2.1 assuming a probability level $\alpha = 0.95$, and use five-minute high-frequency intraday returns to identify the daily jump risk for each subsector. Fig. 1 shows the paths (i.e., the daily logarithm of closing prices) and jumping points for five financial sectors. It can be observed that all of these sectors experienced considerable jumps during the sample period. In particular, they experienced the biggest downside jump on February 3, 2020, the first trading day following the Chinese Lunar New Year holiday, which coincided with the outbreak of COVID-19 pandemic, which affected all indices. More descriptive statistics for jump frequencies and pairwise jumps plots can be found in the Appendix A. These results reveal numerous asymmetries in jump frequency and size, as well as pairwise jumps in financial subsectors.

We then show the results of the correlation analysis in Table 2. In this table, the upper two matrices correspond to the correlation coefficients for the returns and BJ_t, respectively, of the five financial subsectors. The results of these two matrices reflect that there are significant co-movements in the returns of and bilateral jumps between five sub-sectors. Specifically, the securities and diversified financial sectors exhibit the highest correlation of returns of 0.764 and the real estate and securities sectors have the highest BJ_t correlation of 0.602, while the diversified finance and banking sectors have the lowest BJ_t correlation and correlation of returns of 0.468 and 0.303, respectively. Subsequently, we decompose BJ_t into PJ_t and NJ_t to compare the differences between upside and downside jump risks. From the remaining two matrices, we can clearly observe that the correlations of NJ_t are almost twice those of the corresponding of the PJ_t, These results reflect that downside jump risk is more systemic in the five financial subsectors, while the upside jumps are more idiosyncratic. Therefore, in practice, we should not blur the distinction between upside and downside jump risks, but rather analyze them in a disaggregated manner.

[1] These categories are derived from the CSRC Industry Classification Guidelines (http://www.csrc.gov.cn/csrc/c101864/c1024632/content.shtml). Since the real estate sector plays an important role in China's financial system, we include it in our financial risk analysis.

Fig. 1. Sector indices and jumps.

3.2. *Jump risk contagion analysis*

Section 3.1 suggests that a jump in one sector is often accompanied by co-jumps in other sectors. This seems to imply that there is a contagion effect of jump risk among the financial sectors. Recently, the contagion of financial risk is of increasing concern to risk managers and policymakers. In practice, both need to dynamically identify the roles played by different nodes in the risk network system and determine whether each node is a risk contributor or a risk taker such that they can take timely action to prevent risk shocks or maintain the stability of the financial system. However, cross-correlation analysis only measures pairwise nondirectional association and thus depends on linear thinking. To provide deeper insight into the dynamic features of jumps, we use a network approach in this section to investigate the topology of jump risk contagion. Referring to Section 2.2, we separately introduce BJ_t, PJ_t, and NJ_t into this framework to analyze the differences between various jump risk contagion topologies. Meanwhile, we compute the total connectedness to measure various risk contagion levels across the five sample subsectors.

3.2.1. *Full-sample static analysis*

Table 3 reports the full-sample static jump risk contagion connection. The connectedness relies on an optimal VAR model selected with 1-order lag, and 5-day ahead forecast errors variance decompositions.

Panel A of Table 3 takes the connectedness of the full-sample static bilateral jump risk as a benchmark. Several features deserve our attention. First, that the diagonal elements (i.e., self-contagion) are often the largest in the upper-left block reflects that the primary

Table 2
Correlation analysis results.

	Real estate	Diversified finance	Securities	Banking	Insurance	Real estate	Diversified finance	Securities	Banking	Insurance
	5 min return					Bilateral jump variation (BJ_t)				
Real estate	1.000***					1.000***				
Diversified finance	0.621***	1.000***				0.531***	1.000***			
Securities	0.625***	0.764***	1.000***			0.602***	0.560***	1.000***		
Banking	0.537***	0.468***	0.490***	1.000***		0.406***	0.303***	0.360***	1.000***	
Insurance	0.618***	0.556***	0.624***	0.587***	1.000***	0.481***	0.411***	0.466***	0.373***	1.000***
	upside jump variation (PJ_t)					downside jump variation (NJ_t)				
Real estate	1.000***					1.000***				
Diversified finance	0.240***	1.000***				0.681***	1.000***			
Securities	0.380***	0.340***	1.000***			0.759***	0.758***	1.000***		
Banking	0.295***	0.190***	0.221***	1.000***		0.457***	0.333***	0.461***	1.000***	
Insurance	0.262***	0.251***	0.342***	0.458***	1.000***	0.644***	0.530***	0.588***	0.379***	1.000***

Note: This table shows the cross-correlation of five-minute returns and various jump variations for five subsectors, including the real estate, diversified finance, securities, banking, and insurance sectors. The sample period is from December 13, 2018, to July 23, 2021. The top two matrices correspond to the cross-correlation of five-minute returns and BJ_t, respectively, and the bottom two matrices show the cross-correlation of PJ_t and NJ_t, respectively. We calculate correlation using the Pearson coefficient method. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3
Full-sample static jump risk contagion.

Panel A: Bilateral jump risk contagion						
	Real estate	Securities	Diversified finance	Insurance	Banking	**From others**
Real estate	48.599	18.192	13.894	11.370	7.944	51.401
Securities	17.880	48.355	16.533	10.638	6.594	51.645
Diversity finance	15.125	17.913	52.761	9.237	4.964	47.239
Insurance	13.221	12.424	9.678	56.733	7.944	43.267
Banking	10.662	8.840	6.116	9.165	65.218	34.782
To others	56.888	57.369	46.221	40.410	27.447	**45.667**
Net contribution	5.487	5.724	−1.018	−2.857	−7.335	
Panel B: Upside jump risk contagion						
	Real estate	Securities	Diversified finance	Insurance	Banking	**From others**
Real estate	73.017	10.695	4.899	4.865	6.524	26.983
Securities	9.811	69.027	9.627	8.073	3.463	30.973
Diversified finance	5.139	10.814	75.717	4.899	3.431	24.283
Insurance	4.534	7.979	4.472	68.364	14.651	31.636
Banking	6.037	3.577	3.068	15.349	71.968	28.032
To others	25.521	33.066	22.066	33.186	28.069	**28.381**
Net contribution	−1.462	2.093	−2.217	1.550	0.037	
Panel C: Downside jump risk contagion						
	Real estate	Securities	Diversified finance	Insurance	Banking	**From others**
Real estate	37.767	21.560	17.167	15.688	7.818	62.233
Securities	21.178	37.109	21.033	12.806	7.874	62.891
Diversified finance	18.853	23.495	41.521	11.622	4.508	58.479
Insurance	19.104	15.719	12.784	45.809	6.584	54.191
Banking	12.391	12.723	6.526	8.579	59.782	40.218
To others	71.527	73.497	57.510	48.695	26.784	**55.603**
Net contribution	9.294	10.606	−0.969	−5.496	−13.434	

Note: This table reports the full-sample static jump risk contagion connectedness. The connectedness relies on an optimal VAR model selected with 1-order lag and 5-day-ahead forecast error variance decompositions. The upper-left 5 × 5 block in each Panel shows the results of generalized variance decompositions. The rightmost column gives the total directional connectedness (i.e., the contribution from all others to *i*); The penultimate row gives the total directional connectedness (i.e., the contribution to all others from *j*); the bottom-most row gives the difference in total directional connectedness (i.e., the net contribution to all others). The bottom-right element is the total connectedness.

source of bilateral jump risk in any sector is endogenous. Second, the directional contagions (from or to others) tend to be higher in the real estate and securities sectors than their respective diagonal elements, while the diversified finance, insurance, and banking sectors exhibit the opposite phenomenon. Third, bilateral jump risks interacted across financial subsectors with the total connectedness over the whole sample period reach 45.667. Last, in the "Net contribution" row, it can be seen that the securities sector (5.724) and the real estate sector (5.487) are the top two net contributors to bilateral jump risk contagion in the financial sector as a whole, while the banking sector (−7.335) is the primary net recipient, followed by the insurance sector(−2.857) and the diversified finance sector (−1.018).

We note that the results for bilateral jump risk contagion shown in Panel A of Table 3 do not affect the distinctions between upside and downside risk contagion. To date, many scholars have found significant asymmetries between upside and downside risks. Patton and Sheppard (2015) show that the impact of a price jump on volatility depends on the sign of the jump, with negative (positive) jumps leading to higher (lower) future fluctuations. Baruník et al. (2016) provide empirical evidence that realized semivariance spillovers exhibit asymmetries. Therefore, we explore whether there is also an asymmetric effect for contagion between upside and downside jump risks. To do so, we deconstruct BJ_t into PJ_t and NJ_t to separately analyze upside and downside jump risk contagion.

Panel B of Table 3 provides the detailed full-sample directional connectedness for upside jump risk among financial sectors. We note that it is similar to Panel A in that the elements on the diagonal are also the largest in the upper-left decompositions but with relatively higher levels. However, unlike Panel A, on the one hand, all total directional contagions (from others or to others) in Panel B are reduced. On the other hand, the total connectedness decreases sharply to only 28.381.

Such results suggest that own-effects (i.e., the diagonal elements) perform more strongly while the interaction effects perform more weakly in the upside jump risk network. Furthermore, it can be seen from the "Net contribution" row that some changes are observed. Compared to the corresponding in Panel A, the securities sector (2.093) continues to play a leading role in net contribution to the others, followed by the insurance sector (1.550) and the banking sector (0.037). However, the evidence indicates that the performances of both the banking and insurance sectors shift from being net recipients in the bilateral jump risk network to being net contributors in the upside jump risk network. In addition, the diversified finance sector (−2.217) and the real estate sector (−1.462) are both two net recipients.

Similarly, we display the full-sample downside jump risk network in Panel C of Table 3. A few salient changes are worth noting when comparing Panel C with the first two Panels. For instance, although the elements on the diagonal remain the largest in the left-upper matrix, the values exhibit a noticeable decline. This implies less of a self-contagion effect and more of a cross-contagion effect for downside jump risk. In addition, all total directional contagions (from or to others) in Panel C tend to be much larger. Furthermore, the total connectedness rises sharply to 55.603, which is almost twice the total upside jump risk connectedness (28.381) and exceeds the total bilateral jump risk connectedness (45.667) by 9.936, which implies that downside jump risk is more contagious. Last, we shed light on the asymmetries in the "Net contribution" between these three Panels. As with the rankings in Panel A in Table 3, all subsectors play the same roles, but with different intensities, in the network for downside jump risk. However, when we compare Panel C with Panel B, the results in the last row highlight additional asymmetric effects. First, the securities sector continues to act as a net contributor, but its net contribution to other subsectors in the downside jump risk network is significantly larger than that in the upside jump risk network (i.e., 10.606 vs. 2.093). Second, the net recipients in the diversified finance sector are weaker in the downside jump risk network than in the upside jump risk network (−0.969 vs. −2.217). In addition, the roles of the banking, insurance, and real estate sectors have switched. The banking sector changes from being a net contributor (0.037) to being a net recipient (−13.434), accompanying the insurance sector in changing from being a net contributor (1.550) to a net recipient (−5.496) while the real estate sector changes from being a net recipient (−1.462) to a net contributor (9.294).

In summary, the full-sample analysis illuminates the complexity of the various jump risk contagion networks and captures the heterogeneity between the downside and the upside jump risk networks. This evidence implies that differentiating between upside and downside contagion is necessary for risk managers to fully understand the network structures of extreme risk contagions, as several financial subsectors assume different roles in different risk networks and have different contagion effects.

In Fig. 2, we also use chord diagrams to visualize the various jump risk contagion topologies among financial subsectors, in which the three subplots correspond to Panel A, B, and C in Table 3, respectively. In each subplot, we mark the sample sectors with five colors, namely, real estate in blue, securities in red, diversified finance in green, insurance in purple and banking in brown. They are arranged radially around the outer side of a circle. The directional interactions between the sectors are drawn as arc ribbons connecting the sectors. It can be observed that each subsector corresponds to the eight colored arc ribbons of the inner circle, and the four arc ribbons with the same color as the outer circle indicate the risk outputs, while the remaining four arc ribbons with different colors from the outer circle indicate the risk inputs. Specifically, those colored arc ribbons match the off-diagonal elements of the 5×5 matrix of Table 3 (20 in total), while the thickness of each arc ribbon also corresponds to its directional connectedness.

The three subplots in Fig. 2 characterize the cross-sectoral contagions for various financial jump risks. Specifically, the top two subplots display the structural differences between upside and downside jump risk contagion. For instance, the thickest purple and brown arc ribbons bridging the banking and insurance sectors in the upper-left subplot indicate a strong upside jump risk contagion between them. However, in the upper-right subplot, the thickest red arc ribbon reflects that there is more downside jump contagion from the securities sector to the diversified finance sector. In addition, the bilateral jump risk contagion topology appears notably similar to the downside jump risk contagion. The slight differences between them are that the bottom subplot shows that the most prolific external source of bilateral jump risk for the banking sector is the blue arc ribbon (i.e., the real estate sector), followed by the purple, red, and green ribbons. However, the upper-right subplot shows that the most prolific source from others in the downside jump risk network to the banking sector is the red arc ribbon (i.e., the securities sector), followed by the blue, purple, and green ribbons.

3.2.2. Rolling-sample dynamic analysis

Financial markets are changing rapidly, yet full-sample static analysis can only capture time-invariant characteristics without detecting the precise cross-sectional mechanism of jump risk transmission. Therefore, the dynamic analysis results may be more practical for financial risk managers and policymakers. To this end, we perform a dynamic estimation on the 90-day rolling samples and assess the risk connections for the bilateral, upside, and downside jumps across sectors over time.

Fig. 3 shows the total dynamic connectedness for each jump risk corresponding to the right-bottom element ($Cidx^H$) in Table 1. We can observe that there is time-variant asymmetries in various jump risk contagions. For example, starting from the second half of 2019 and continuing until February 2020, the total connectedness of all three jump risks though slowly increase and then decrease. However, during this period, the total connectedness of bilateral and upside jump risk increase from around 30 to over 60, while the total connectedness of downside jump risk increases from 50 to nearly 70. They then trend downward and bottom out around the end of the first quarter of 2020. Also, the total connectedness for both bilateral and downside jump risk rises sharply to above 70 after the COVID-19 pandemic outbreak, which continues until the end of September 2020. However, the total connectedness for upside jump risk rises slowly to a peak of 60. In fact, the difference between downside and upside jump risk contagions during this period is not difficult to explain. This is mainly due to the fact that the sudden outbreak of the pandemic strongly shocked the Chinese economic system, with short-term ripple effects across the financial system leading to a systemic collapse of different stock indices. As a result, the risk of downside jump contagion rose. Subsequently, the Chinese government took aggressive anti-pandemic and economic stimulus actions. As the economic environment gradually improved, investment sentiment also turned positive. Naturally, market sentiment improved and upside risk contagion rose across the financial sector. In the following year, the COVID-19 outbreak spread globally, and the global shutdown fed back into the Chinese economic system. As such, it can be seen in Fig. 3 that downside and upside risk contagion experience reversals between September 2020 and October 2021, with upside risk contagion declining and downside risk contagion rising.

After the Chinese National Day holiday in 2020, the total connectedness of bilateral or upside jump risk drops and remains low at around 30 to 40 until the first half of 2021, while the total connectedness of downside jump risk experiences a sharp rise and fall in the same period. Last, the total connectedness of bilateral jump risk has remained at 30 since the latter half of 2021. By contrast, the total connectedness of upside jump risk experienced a slight increase and then fall, while the total connectedness of downside jump risk shows the opposite.

In addition, we also calculated the mean of each contagion index. We found that the average contagion of bilateral, upside, and downside jump risks over the sample period are 44.634, 39.839, and 47.524, respectively. This result seems to similarly imply that downside jump risk is more contagious than upside jump risk. To compare the difference in

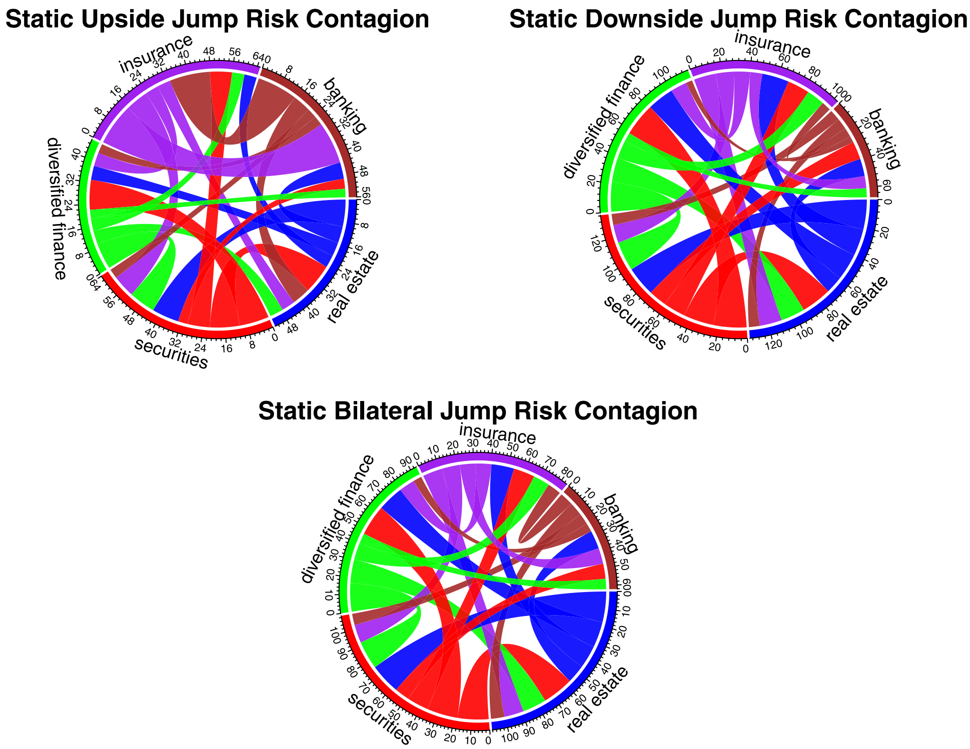

Fig. 2. Full-sample static jump risk contagion topology structures.

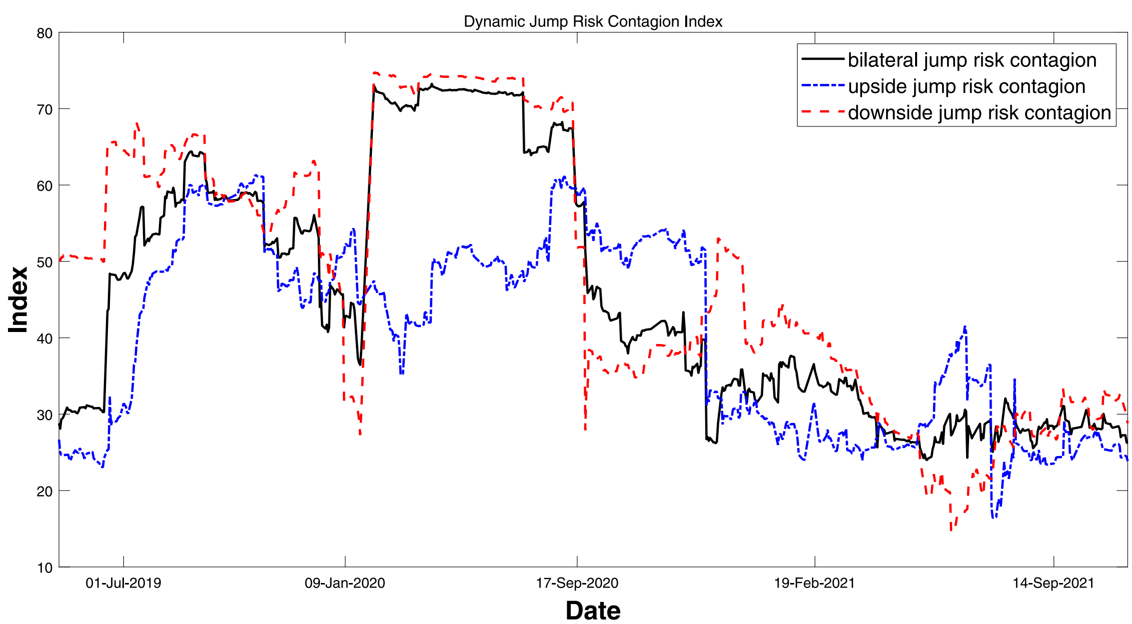

Fig. 3. Rolling-sample dynamic jump risk contagion.

total contagion between upside and downside jump risk at the average level, we perform a one-tailed hypothesis test using the t-statistic. The results support that conclusion at the 5% significance level with a p value of 0.025.

In Appendix B, we provide further analysis concerning the net directional connectedness as well as the dynamic topologies for various jump risks. The figures provide us with more precise information about the dynamic jump risk contagion structures and reveal additional complexities and asymmetries.

3.3. Explaining the asymmetry of jump risk contagion

Recognizing the driving mechanisms behind the jump risk contagion can be useful for risk managers and policymakers. For example,

Table 4
Descriptive statistics of variables.

	Mean	Median	Maximum	Minimum	Std.dev.	Skewness	Kurtosis
Bi_Cidx	44.634	40.452	72.431	25.695	16.128	0.524	1.811
Up_Cidx	39.839	41.800	59.830	24.608	12.410	0.070	1.443
Dn_Cidx	47.524	42.768	74.164	18.709	17.430	0.155	1.706
$M2$	9.258	8.700	11.100	8.100	1.092	0.557	1.665
$AFRE$	11.618	11.025	13.700	10.000	1.229	0.406	1.674
$PMPT$	−2.360	−2.400	−0.900	−3.600	0.686	0.187	2.443
EPU	672.816	665.308	970.830	358.358	192.041	−0.019	1.755
PB	1.990	1.899	2.970	1.472	0.405	1.129	3.501
$TURNOVER$	161.956	150.320	413.760	71.170	75.486	1.339	5.111
$CFCI$	−0.818	−0.819	−0.277	−1.448	0.261	−0.318	3.174

Note: This table shows the descriptive statistics of explained and explanatory variables. Bi_Cidx, Up_Cidx, and Dn_Cidx indicate the total connectedness ($Cidx$) for bilateral, upside, and downside jump risk, respectively. *M2* and *AFRE* indicate the growth rates of M2 and aggregate financing to the real economy, respectively, and their units are both denominated in %. *PMPT* indicates the degree of poor monetary policy transmission, which is the difference between *M2* and *AFRE*. *EPU*, *PB*, and *CFCI* indicate economic policy uncertainty, the average PB ratio of five sample sectors, and the China Financial Conditions Index, respectively, and all three variables are unitless. *TURNOVER* indicates the monthly average market turnover of the five sample sectors, and its unit is 100 million Yuan.

risk managers can optimize an early risk warning indicator system from the perspective of risk contagion to improve the efficiency of real-time warning and prevent systemic shocks and extreme risks. In addition, national economic policy authorities can recognize the effects the policies they implement have on financial markets, which helps them optimize policy formulation and maintain economic and financial market stability.

There has been a recent increase in the study of the drivers behind financial risk contagion. It is well documented that the COVID-19 pandemic, economic policy uncertainty, industry-specific characteristics, quantitative easing, and stock market liquidity not only induce financial risk but also play a role in its contagion of financial risk such that Bernal et al. (2016), Guo et al. (2021), Iwanicz-Drozdowska et al. (2021), Lien et al. (2022). Although these studies contribute to our understanding of the mechanisms influencing risk contagion in financial markets, they all ignore the distinction between upside and downside risk, and thus none of them investigate the mechanisms behind the asymmetric effects of risk contagion. In the following subsection, we analyze the asymmetric effects of jump risk contagion from the perspective of monetary policy transmission.

3.3.1. The role of monetary policy transmission

The main purpose of monetary policy is to regulate the functioning of the real economy. However, inefficient monetary policy transmission is often responsible for monetary policy failures. More importantly, when the central bank adopts a loose monetary policy, if the real sector lacks the willingness to raise funds, poor monetary policy transmission often leads to a large amount of money stagnating in the financial sector instead of being injected into the real sector to support economic activity. The idling of funds in the financial sector raises financial leverage and promotes excessive speculative activities, which will result in the formation of financial bubbles. It is worth noting that financial bubbles can form slowly yet burst systematically, thus creating very different characteristics in upside and downside risk connectedness. Once an unfavorable extreme risk event occurs in a financial subsector, the entire financial system will form contagion through the financial risk network and a systemic crisis will follow. Based on this, we empirically test the asymmetric impact of monetary policy transmission on the jump risk contagion in the Chinese financial sector.

In the empirical regression, we consider that the monetary policy transmission is mainly influenced by two forces: central bank money supply and real economy's aggregate financing.[2] Therefore, the difference between the growth rate of central bank money supply (*M2*) and the total financing of the real economy (*AFRE*), denoted as *PMPT*, can be used to reflect the degree of poor monetary policy transmission (i.e., an increase in *PMPT* implies a weakening of monetary policy transmission and raises the propensity for funds to remain idle in the financial sector and vice versa). Moreover, we control for the effects of several other potential drivers of jump risk contagion. These factors include economic policy uncertainty (*EPU*), financial sector valuation (*PB*), market turnover (*TURNOVER*), and Chinese financial conditions (*CFCI*). See Appendix C for additional details on the definitions and sources of these variables. Moreover, we use the dynamic total connectedness ($Cidx$) for bilateral, upside, and downside jump risks as explained variables and relabel them as Bi_Cidx, Up_Cidx, and Dn_Cidx, respectively. Since several of the above variables use daily data, we regularize them into monthly frequencies by calculating simple averages for each month.

Table 4 provides descriptive statistics for each variable. The mean values of the various dynamic total connectivity measures Bi_Cidx, Up_Cidx and Dn_Cidx show that in our sample period, downside jump risk is most contagious, followed by bilateral and upward jump risk. The average growth rate of *AFRE* is higher than that of *M2*. *PMPT* ranges from −2.400 to −0.900 with a negative mean of −2.360. In addition, the mean value of *EPU* is 672.816, with negative skewness. The kurtosis of *PB*, *TURNOVER*, and *CFCI* are all relatively high at 3.501, 5.111, and 3.174, respectively.

Subsequently, Table 5 exhibits the results of correlation analysis. Some intuitive conclusions can be derived from the first three columns. First, the correlation between $PMPT$ and Bi_Cidx is 0.459 at the 1% significance level, which is higher than the correlation coefficient (0.390) between $M2$ and Bi_Cidx. This thus implies that, compared to the money supply, monetary policy transmission has a greater impact on jump risk contagion. Furthermore, the correlation coefficient between $PMPT$ and Dn_Cidx is 0.363 at the 5% significance level, while that between $PMPT$ and Up_Cidx is only 0.189 and nonsignificant. This result implies that downside jump risk contagion is more sensitive to monetary policy transmission. Second, EPU has the highest correlation with the three contagion indices Bi_Cidx, Up_Cidx, and Dn_Cidx, reaching 0.759, 0.644, and 0.836, respectively. Third, the correlation coefficients between PB and Bi_Cidx or Dn_Cidx are both negative at the 5% significance level, reaching −0.417 and −0.453, respectively, but the correlation between PB and Up_Cidx is only 0.067 and nonsignificant. In addition, the remaining seven columns show the correlation between the explanatory variables. Among them, the highest correlation between $M2$ and $AFRE$ is 0.831, while $TURNOVER$ and $CFCI$ have the lowest correlation at −0.004.

Fig. 4 shows the trend and scatter plots of $PMPT$ versus the three contagion indices. In subplot (a), we can observe the synergistic movement characteristics between $PMPT$ and the three contagion indices, and the trend of $PMPT$ is graphically more similar to those

[2] The Chinese central bank's monetary policy framework is a combination of quantitative and price-based policy regulation, with quantitative regulation dominating. Therefore, this study focuses on the role of quantitative monetary policy transmission.

Table 5
The correlation of variables.

	Bi_Cidx	Up_Cidx	Dn_Cidx	$M2$	$AFRE$	$PMPT$	EPU	PB	$TURNOVER$	$CFCI$
$M2$	0.390**	0.381**	0.335*	1.000***	0.831***	0.103	0.203	0.399**	0.357**	−0.410**
$AFRE$	0.090	0.233	0.095	0.831***	1.000***	−0.467***	0.086	0.600***	0.479***	−0.081
$PMPT$	0.459***	0.189	0.363**	0.103	−0.467***	1.000***	0.169	−0.439**	−0.290	−0.507***
EPU	0.759***	0.644***	0.836***	0.203	0.086	0.169	1.000***	−0.343*	−0.371**	0.242
PB	−0.417**	0.067	−0.453**	0.399**	0.600***	−0.439**	−0.343*	1.000***	0.543***	0.048
$TURNOVER$	−0.227	−0.089	−0.207	0.357**	0.479***	−0.290	−0.371**	0.543***	1.000***	−0.004
$CFCI$	−0.152	0.006	0.064	−0.410**	−0.081	−0.507***	0.242	0.048	−0.004	1.000***

Note: Bi_Cidx, Up_Cidx, and Dn_Cidx indicate the total connectedness ($Cidx$) for bilateral, upside, and downside jump risk, respectively. *M2* and *AFRE* indicate the growth rates of M2 and aggregate financing to the real economy, respectively, and their units are both denominated in %. *PMPT* indicates the degree of poor monetary policy transmission, which is the difference between M2 and AFRE. *EPU*, *PB*, and *CFCI* indicate economic policy uncertainty, the average PB ratio of five sample sectors, and the China Financial Conditions Index, respectively, and all three variables are unitless. *TURNOVER* indicates the monthly average market turnover of the five sample sectors, and its unit is 100 million Yuan. Symbols ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

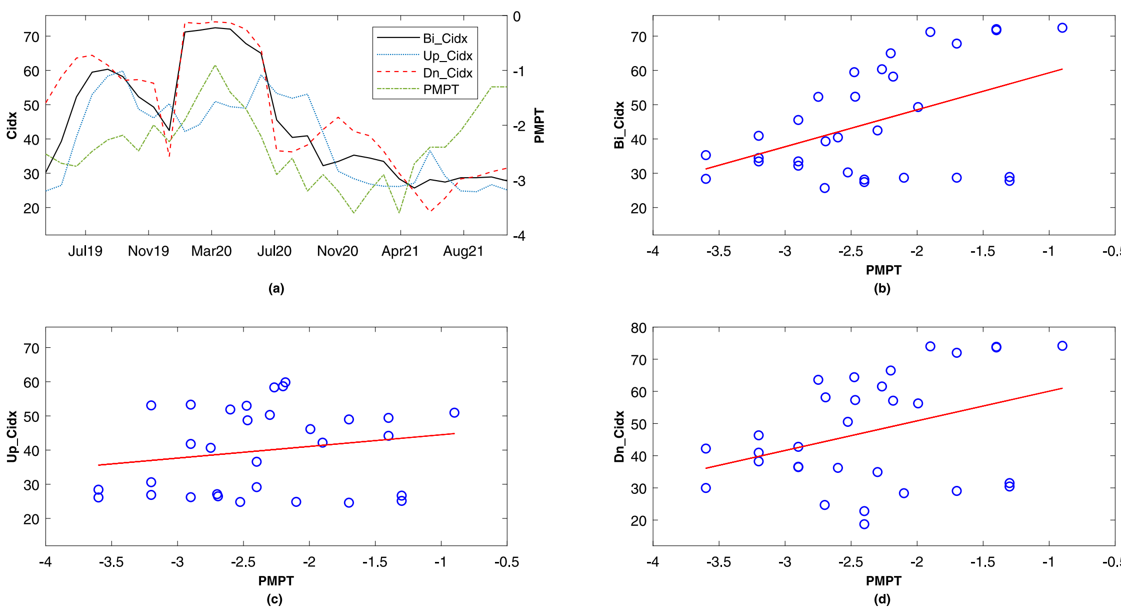

Fig. 4. Monetary policy transmission & jump risk contagion.

of Bi_Cidx and Dn_Cidx. The remaining three subplots (b)–(d) show scatter plots and univariate OLS-fitted lines between $PMPT$ and the three contagion indices, respectively. These three plots also intuitively reflect the positive correlation characteristic between monetary policy transmission and jump risk contagion. In addition, it can be seen from the slopes of the three univariate OLS fitted lines that monetary policy transmission is more correlated with downside jump risk contagion.

We then empirically investigate the impacts of these potential drivers on financial jump risk contagion. OLS regression using level variables can cause pseudo-regression problems.[3] Therefore, it is appropriate to construct error correction models using cointegrated variables.[4] Subsequently, we analyze three specifications for the contagions of bilateral, upside, and downside jump risks, respectively. The uniform specification is given by

$$\Delta Cidx_t = \boldsymbol{\beta}' \Delta \boldsymbol{X}_t + \gamma Covid19_t + \lambda ECM_{t-1} + \varepsilon_t \tag{3.1}$$

where Δ is the difference operator; $Cidx$ can be replaced by total connectedness Bi_Cidx, Up_Cidx, or Dn_Cidx; and $\boldsymbol{\beta}$ is a coefficient vector corresponding to the explanatory variable vector $\boldsymbol{X}$. We also control the impact of the COVID-19 pandemic outbreak by incorporating the dummy variable $Covid19$.[5] In addition, ECM_{t-1} is the error correction term and ε_t is the disturbance term.

Table 6 shows the regression results for bilateral risk contagion. Columns (1)-(7) show the univariate regression results after controlling for the effect of the COVID-19 pandemic. It can be observed that the coefficient of ΔEPU in column (1) is significantly positive at 0.045, while the coefficient of $Covid19$ is significantly positive at 4.565. Meanwhile, the adjusted R^2 also reaches 0.387. The results of column (1) suggest that the elevated economic policy uncertainty and the outbreak of the pandemic are indeed key factors triggering the cross-sectoral contagion of jump risk, which is similar to the findings of Lien et al. (2022)

[3] The unit root test results for these variables are attached in Appendix D The ADF test results show that all level variables accept the null hypothesis of the existence of at least one unit root at the 1% significance level. Moreover, all first-order difference variables reject the null hypothesis, thus implying that these variables are all I(1) series.

[4] The Johansen cointegration test results for our variables are also listed in Appendix E and indicate these alternative variables sets all accept the null hypothesis of the existence of at most three cointegrating equations at the 0.05 level.

[5] Although China implemented national emergency response measures for the Covid-19 outbreak at the end of January 2020, it was during the Chinese Lunar New Year holiday, when the stock market was closed, and the stock market reopened on February 3, 2020. Therefore, we construct the COVID-19 dummy variable as

$$Covid19_t = \begin{cases} 0 & , \quad if\ t < February\ 2020 \\ 1 & , \quad if\ t \geq February\ 2020 \end{cases} .$$

Table 6
Regression results for bilateral jump risk contagion.

	Dependent variable: ΔBi_Cidx									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ΔEPU	0.045***							0.040***	0.040***	0.039***
	(4.041)							(4.955)	(5.855)	(5.691)
ΔPB		−9.192						−5.056	−3.952	−4.374
		(−1.501)						(−1.140)	(−0.996)	(−1.162)
$\Delta TURNOVER$			−0.021					0.002	−0.007	−0.002
			(−1.186)					(0.187)	(−0.674)	(−0.225)
$\Delta CFCI$				−6.312				5.920	14.949**	12.249**
				(−0.696)				(0.936)	(2.608)	(2.178)
$\Delta M2$					2.350				4.394**	
					(0.896)				(2.413)	
$\Delta AFRE$						0.964			−7.556**	
						(0.240)			(−2.872)	
$\Delta PMPT$							5.002			4.844**
							(1.522)			(2.668)
$Covid19$	4.565**	−0.090	−0.797	0.221	−1.056	−0.942	−3.458	13.527***	21.561***	17.719***
	(2.473)	(−0.062)	(−0.494)	(0.129)	(−0.661)	(−0.588)	(−1.551)	(5.784)	(7.597)	(7.342)
ECM(−1)	−0.506***	−0.180***	−0.144*	−0.093*	−0.101*	−0.127*	−0.160	−0.531***	−0.543***	−0.599***
	(−3.842)	(−3.162)	(−1.928)	(−1.728)	(−2.050)	(−2.040)	(−1.621)	(−6.584)	(−8.179)	(−8.095)
R^2	0.430	0.314	0.131	0.146	0.159	0.149	0.170	0.708	0.809	0.800
$Adj.R^2$	0.387	0.264	0.067	0.083	0.097	0.086	0.108	0.648	0.748	0.747

Note: T-statistics are in parentheses, symbols ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

and Cao (2022). The nonsignificance of the coefficients of columns (2)–(7) is mainly due to the omission of important variables. Therefore, we further consider the regression results of column (8), which can be used as a benchmark. The regression results of this column likewise suggest that elevated economic policy uncertainty and the outbreak of the COVID-19 pandemic are key factors that increase jump risk contagion. Moreover, the multivariate regression improves the explanatory power of the model, with R^2 and adjusted R^2 reaching 0.708 and 0.648, respectively.

Subsequently, when $\Delta M2$ and $\Delta AFRE$ are added to regression (9) of Table 6, it can be observed that the interpretation of the model is improved, but their effects on risk contagion are significantly opposed, with coefficients of 4.394 and −7.556. Moreover, the coefficient of $\Delta CFCI$ is significant at the 1% level of 14.949, which implies that an increase in the financial conditions index (i.e., the deterioration of the financing environment) promotes the cross-sectoral contagion of bilateral jump risk. Consistent with the results of regression (8), elevated economic policy uncertainty and the COVID-19 pandemic outbreak significantly increase bilateral jump risk contagion at the 1% confidence level, with regression coefficients of 0.040 and 21.561, respectively.

In addition, both ΔPB and $\Delta TURNOVER$ have nonsignificant coefficients in column (9) of Table 6. Changes in valuations throughout the financial sector and stock market turnover seem not to cause bilateral jump risk contagion in the short term. Furthermore, the coefficient of the error correction term $ECM(-1)$ is also significant at the 1% confidence level of −0.543, with its value greater than −1 and less than 0 implying the existence of a long-run equilibrium relationship between these variables. The R^2 and adjusted R^2 of this regression achieve 0.809 and 0.748, respectively, which are both higher than the corresponding results in column (8). Also, we can derive similar conclusions to those in column (9) when we introduce $\Delta PMPT$ into the regression. The results in column (10) suggest that an increase in $\Delta PMPT$ can promote cross-sectional contagion of jump risk significantly given its coefficient of 4.844 at the 5% significance level. The significance of the other coefficients is similar to those in column (9). Meanwhile, the R^2 and adjusted R^2 of the model are 0.800 and 0.747, respectively.

In summary, the results in Table 6 suggest that economic policy uncertainty, the COVID-19 pandemic, industry valuation, financial conditions, money supply, and aggregate financing to the real economy are drivers of jump risk contagion. The impact of money supply and aggregate financing to the real economy on jump risk contagion implies that monetary policy transmission plays an important role in and poor monetary policy transmission promotes jump risk contagion.

Although the empirical evidence presented above is convincing, the regression analysis is poor in that it blurs the distinction between upside and downside risk contagion. As discussed in Section 3.2, there are many asymmetries in upside and downside jump risk contagions. It is logical to question what causes the asymmetric contagions in the two jump risks. To answer this question, we focus on the role played by monetary policy transmission in this context.

Table 7 reports the regression results for jump risk asymmetric contagion. The explanatory variables for each regression in Panel A are the upside jump risk contagion index(Up_Cidx), while all explanatory variables in Panel B are the upside jump risk contagion index(Dn_Cidx). First, the results from the univariate regressions after controlling for the COVID-19 pandemic outbreak in columns (1)-(7) roughly show that the COVID-19 pandemic, industry valuation, and economic policy uncertainty are the main factors affecting upward risk contagion at the 5% significance level. Then, we use the multivariate regression model (8) as a benchmark. The results in column (8) show that the coefficients of ΔEPU and ΔPB are significantly positive at the 5% significance level at 0.019 and 16.958, respectively, while the coefficients of $\Delta CFCI$ and $Covid19$ are significantly negative at −12.502 and −7.724, respectively. The regression R^2 and adjusted R^2 are 0.649 and 0.576, respectively. These results imply that after controlling for other factors, an increase in economic policy uncertainty or industry valuations can boost the cross-sectional contagion of upside jump risk, while the deterioration of the financing environment and the outbreak of the COVID-19 pandemic reduce it. In regression (9) we add $\Delta M2$ and $\Delta AFRE$-two variables associated with monetary policy transmission-and find that neither of them is significant at the 5% significance level. This suggests that changes in the money supply or aggregate financing to the real economy have no effect on upside risk contagion at the 5% significance level, which reflects that monetary policy plays no role in the cross-sectoral contagion of upside jump risk. Compared to column (8), the original coefficients change only slightly with the significance level remaining unchanged, and the same is true for the R^2 and adjusted R^2. Furthermore, we introduce $\Delta PMPT$ into the regression model, and the results in column (10) show that monetary policy transmission does not affect the cross-sectional contagion of upside jump risk.

Table 7
Regression results in the jump risk asymmetric contagion.

	Panel A: *ΔUp_Cidx* as dependent variable									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ΔEPU	0.020**							0.019**	0.017**	0.019**
	(2.359)							(2.626)	(2.303)	(2.562)
ΔPB		13.223***						16.958***	16.376***	16.943***
		(2.904)						(4.211)	(3.724)	(4.120)
ΔTURNOVER			−0.024*					0.001	0.001	0.000
			(−1.836)					(0.054)	(0.131)	(0.020)
ΔCFCI				−13.109*				−12.502**	−12.933**	−12.527**
				(−2.022)				(−2.298)	(−2.223)	(−2.190)
ΔM2					3.971*				0.678	
					(1.896)				(0.335)	
ΔAFRE						5.795*			0.372	
						(1.789)			(0.126)	
$\Delta PMPT$							0.761			0.180
							(0.297)			(0.091)
Covid19	−1.332	−5.197***	−2.324*	−9.204***	−9.722***	−6.716***	−5.704***	−7.724***	−9.595***	−7.793***
	(−1.205)	(−3.553)	(−1.862)	(−3.060)	(−3.443)	(−3.093)	(−2.878)	(−4.920)	(−4.870)	(−4.826)
ECM(−1)	−0.370***	−0.224***	−0.151***	−0.187***	−0.329***	−0.290***	−0.213***	−0.422***	−0.418***	−0.422***
	(−4.052)	(−3.828)	(−3.209)	(−2.862)	(−3.336)	(−3.107)	(−2.888)	(−4.831)	(−4.625)	(−4.720)
R^2	0.406	0.450	0.301	0.314	0.318	0.282	0.263	0.649	0.653	0.650
$Adj.R^2$	0.362	0.409	0.249	0.263	0.267	0.229	0.208	0.576	0.543	0.558
	Panel B: *ΔDn_Cidx* as dependent variable									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ΔEPU	0.059***							0.057***	0.051***	0.051***
	(4.312)							(5.269)	(6.057)	(5.943)
ΔPB		−10.373						−12.448*	−12.121**	−13.060**
		(−1.148)						(−2.031)	(−2.394)	(−2.649)
ΔTURNOVER			−0.002					0.011	−0.004	0.001
			(−0.079)					(0.672)	(−0.331)	(0.105)
ΔCFCI				2.673				16.544*	32.792***	28.572***
				(0.218)				(1.906)	(4.413)	(3.841)
ΔM2					6.021				10.360***	
					(1.669)				(4.395)	
ΔAFRE						3.024			−15.656***	
						(0.544)			(−4.440)	
$\Delta PMPT$							7.772*			8.050***
							(1.757)			(3.371)
Covid19	7.509***	1.643	2.136	3.231	−1.546	0.977	−0.997	21.245***	29.444***	28.060***
	(3.112)	(0.733)	(0.851)	(1.092)	(−0.687)	(0.428)	(−0.464)	(5.695)	(7.807)	(7.431)
ECM(−1)	−0.748***	−0.220**	−0.181*	−0.143*	−0.218*	−0.190*	−0.178	−0.764***	−0.737***	−0.743***
	(−4.274)	(−2.209)	(−1.863)	(−1.741)	(−1.918)	(−1.931)	(−1.535)	(−5.982)	(−8.085)	(−7.721)
R^2	0.476	0.183	0.123	0.098	0.164	0.120	0.181	0.693	0.829	0.810
$Adj.R^2$	0.437	0.123	0.058	0.031	0.102	0.055	0.120	0.630	0.774	0.761

Note: T-statistics are in parentheses, symbols ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

When we focus on the regression results in Panel B, we can draw some meaningful conclusions. The results in columns (1)-(7) also suggest that economic policy uncertainty and the COVID-19 outbreak play roles to some extent, as regression (1) has the strongest explanatory effect among these seven columns with an adjusted R^2 of as high as 0.437. However, unlike the corresponding results in Panel A, the coefficient of $Covid19$ in Panel B is significantly positive at the 1% significance level, thus suggesting that the outbreak of the pandemic increased the cross-sectional downside jump risk contagion. In column (8) of Panel B, the coefficients of ΔEPU, ΔPB, $\Delta CFCI$, and $Covid19$ are significant at the 10% significance level. Compared with column (8) in Panel A, the sign of the coefficients of ΔPB, $\Delta CFCI$, and $Covid19$ changed, and the coefficient of ΔEPU remained positive. Therefore, we can conclude that an increase in industry valuations can reduce downside jump risk contagion, while an increase in economic policy uncertainty, a deterioration in financing conditions, and the outbreak of the COVID-19 pandemic drive downside jump risk contagion.

We are most interested in the results in column (9) of Panel B. when adding $\Delta M2$ and $\Delta AFRE$ to the regression, compared with column (8) of Panel B, it can be seen that the strength of the model is greatly improved, as the t-statistics of the original coefficients become more significant, except that of turnover. More importantly, the coefficients of $\Delta M2$ and $\Delta AFRE$ behave extraordinarily significantly and reach 10.360 and −15.656, respectively, at the 1% significance level, with the t-statistics reaching 4.395 and −4.440. The results for these two coefficients are in sharp contrast to the corresponding results in column (9) of Panel A. This implies that, after controlling for other variables, an increase in the money supply or a decrease in aggregate financing to the real economy drive the cross-sectional downside jump risk contagion. Furthermore, when we focus on the results in column (10) of Panel B, it can be seen that the coefficient of $\Delta PMPT$ is 8.050 at the 1% significance level, while the coefficient of $\Delta PMPT$ corresponding to column (10) of Panel A is 0.180 with a t-statistic of only 0.091. These results explain the asymmetry between the upside and the downside jump risk contagion, and we confirm that monetary policy transmission is an important cause of cross-sectional downside jump risk contagion, yet it does not affect upside risk contagion.

The mechanism behind the above conclusion is not difficult to explain. Weak monetary policy transmission raises the propensity for funds to sit idle in the financial system, while a rapid increase in aggregate financing to the real economy has the opposite effect. Funds' idleness in the financial system promotes financial speculation and asset bubble formation. However, asset bubble formation and bursting

Table 8
Robustness results in the asymmetric effects of monetary policy transmission on jump risk contagion.

	$H = 10$				$p = 2$			
	ΔUp_Cidx		*ΔDn_Cidx*		*ΔUp_Cidx*		*ΔDn_Cidx*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔEPU	0.014* (1.914)	0.017** (2.204)	0.053*** (6.278)	0.053*** (6.089)	0.015* (2.002)	0.017** (2.239)	0.051*** (6.561)	0.051*** (6.386)
ΔPB	16.508*** (3.735)	17.253*** (4.159)	−10.972** (−2.160)	−12.093** (−2.445)	17.825*** (4.077)	18.340*** (4.488)	−11.715** (−2.495)	−12.730** (−2.779)
ΔTURNOVER	0.001 (0.096)	−0.001 (−0.086)	−0.008 (−0.580)	0.000 (0.015)	0.001 (0.066)	−0.000 (−0.044)	−0.006 (−0.472)	0.001 (0.057)
ΔCFCI	−11.959* (−2.049)	−11.136* (−1.932)	34.227*** (4.526)	28.933*** (3.853)	−13.234** (−2.287)	−12.812** (−2.254)	31.672*** (4.606)	27.096*** (3.943)
ΔM2	0.938 (0.462)		9.552*** (4.044)		0.219 (0.109)		9.399*** (4.307)	
ΔAFRE	0.750 (0.254)		−15.752*** (−4.446)		0.797 (0.272)		−14.877*** (−4.540)	
$\Delta PMPT$		0.518 (0.260)		7.904*** (3.296)		−0.199 −(0.101)		7.531*** (3.395)
Covid19	−9.150*** (−4.844)	−6.865*** (−4.663)	30.954*** (7.747)	26.976*** (7.320)	−8.848*** (−4.749)	−7.202*** (−4.701)	29.329*** (8.162)	26.722*** (7.740)
ECM(−1)	−0.430*** (−4.644)	−0.433*** (−4.673)	−0.660*** (−8.008)	−0.718*** (−7.638)	−0.393*** (−4.485)	−0.395*** (−4.589)	−0.723*** (−8.424)	−0.757*** (−8.043)
R^2	0.644	0.638	0.825	0.806	0.657	0.654	0.839	0.822
$Adj.R^2$	0.531	0.543	0.769	0.755	0.548	0.564	0.788	0.775

Note: T-statistics are in parentheses, symbols ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

are asymmetric phenomena. Typically, financial bubbles form idiosyncratically and slowly, but their bursting is systemic in nature. Once an extreme risk event occurs in a particular financial subsector, the risk contagion will form through the financial network and the bursting of the financial bubble will follow, with the increased connectedness of jump risk across the financial sector becoming a concrete manifestation. Therefore, comparing column (9) with Panels A and B, it is clear that $\Delta PMPT$ is actually significant only on Dn_Cidx. In fact, we additionally find exceptionally significant coefficients for ΔPB from columns (8)-(10) in Panel A with t-statistics as high as 4.211, 3.724, and 4.120, while the corresponding t-statistics in columns (8)-(10) of Panel B are only −2.031, −2.394, and −2.649, respectively. It seems that industry valuation has more influence on upside jump risk contagion than does monetary policy transmission. Therefore, we believe that the improvement in the fundamentals of the financial sector is the main factor driving the contagion of upside jump risk.

3.3.2. Robustness check

In this subsection, we test the robustness of the regression results on the asymmetric effects of monetary policy transmission on jump risk contagion by reconstructing the risk contagion indices by changing the parameters outlined in Section 2.2. One approach is to replace the parameter H in Eq. (2.10) with 10 (i.e., to increase the step size of the forecasting). Another way is to adjust the lag order p of the VAR model to 2. We then apply the reconstructed risk contagion indices to the regression analysis. The results of the regressions are displayed in Table 8.

It can be observed that the results in Table 8 are similar to those in Table 7, regardless of whether H is set to 10 or p is set to 2. The regression coefficients in columns (1), (2), (5), and (6) all show that $\Delta M2$, $\Delta AFRE$ and $\Delta PMPT$ have no significant effect on Up_Cidx at the 5% significance level. While the regression results in columns (3), (4), (7), and (8) on Dn_Cidx show that the coefficients of $\Delta M2$, $\Delta AFRE$ and $\Delta PMPT$ are significant at the 1% significance level. These results support that the poor monetary policy transmission drives the downside jump risk contagion, but has no effect on upside jump risk contagion.

4. Conclusions and policy implications

Increasing attention has been paid to the analysis of extreme risk. This paper is one of the few to distinguish between upside and downside jump risk using high-frequency data from the Chinese stock market in a network framework.

In this paper, we focus on the asymmetric characteristics of jump risk contagion in the financial subsectors – namely, the banking, insurance, securities, diversified finance, and real estate sectors – of the Chinese stock market from December 13, 2018, to November 17, 2021. First, we quantify bilateral jump risk by the significant jump variance using five-minute high-frequency data and decompose it into upside and downside jump risk. The correlation results first indicate that downside jump risk is more correlated between financial subsectors than upside jump risk. We then apply Diebold and Yılmaz's spillover framework to separately construct contagion networks and measure the contagions for each jump risk. Subsequently, a static and dynamic network analysis fully demonstrates the complexity of jump risk contagion. In particular, we confirm that downside jump risk is more contagious than upside jump risk and that each financial subsector plays a different role in the various networks. Last, we pay more attention to the mechanism for the asymmetry between upside and downside jump risk contagion throughout the financial sector. Through regression analysis, it is revealed that the poor monetary policy transmission is a significant driver of downside jump risk contagion but has no effect on upside jump risk contagion (i.e., poor monetary policy transmission can explain the asymmetry in jump risk contagion). In addition, we find that economic policy uncertainty, industry valuation, financing conditions, and the COVID-19 pandemic outbreak are also critical to various jump risk contagions across financial subsectors.

There are several policy implications for risk managers and policymakers. Risk managers should not blur the distinction between upside and downside jump risk contagions. Real-time monitoring of both types of jump risk contagion can prevent external risk shocks. Thus, they can incorporate the dynamic contagion indices constructed in our network analysis into their risk management systems to improve their accuracy. Moreover, policymakers should be alert to the impact of poor monetary policy transmission on jump risk contagion in financial markets, especially downside jump risk. This implies that economic monetary policymakers should improve the effectiveness of policy implementation. It is suggested that policymakers should not oversupply financial liquidity with loose policies and should unblock the monetary policy transmission channels so that funds can be injected into the real economy as soon as possible to avoid allowing them to idle in the financial system.

Table A.9
Summary statistics for various jumps.

Panel A: Jump frequency						
	Real estate	Diversified finance	Securities	Banking	Insurance	Mean
# jumps	246	244	215	272	263	248
# positive jumps	111	129	108	117	112	115
# negative jumps	135	115	107	155	151	133
jump frequency	0.347	0.344	0.303	0.384	0.371	0.350
Panel B:Jump risk measures						
	Real estate	Diversified finance	Securities	Banking	Insurance	Mean
average $\lvert BJ_t \rvert$ ($\times$ 10^{-3})	9.465	10.697	12.793	8.676	11.736	10.673
average $\lvert PJ_t \rvert$ ($\times$ 10^{-3})	9.685	10.154	13.708	8.456	11.962	10.793
average $\lvert NJ_t \rvert$ ($\times$ 10^{-3})	9.284	11.306	11.869	8.842	11.569	10.574

Note: Panel A reports jump frequency for five sectors involving the real estate sector, the diversified finance sector, the securities sector, the banking sector, and the insurance sector. The sample period is from December 13, 2018, to July 23, 2021. "# jumps", "# positive jumps", and "# negative jumps" denotes the number of $BJ_t \neq 0$, $PJ_t \neq 0$, and $NJ_t \neq 0$, respectively. "jump frequency" calculates the percentage of $BJ_t \neq 0$ for all sample trading days. Panel B reports the statistics of jump risk measures. To compare sizes, we focus on the absolute values of BJ_t, PJ_t, and NJ_t, respectively. The rightmost column shows the mean values of the five sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Compliance with ethical standards

Ethical approval

This manuscript does not contain any studies with human participants or animals performed by any of the authors.

Informed consent

Informed consent was obtained from all individual participants included in the study.

Appendix A. Descriptive statistics of jump variation

Table A.9 reports some descriptive statistics about the jumps identified in the five sectors. We draw many asymmetric conclusions from this. From the results in Panel A, we can observe that the banking sector has the most intensive jumps, which achieves 272 jumps during the sample period with a jump frequency of 0.384; and the securities sector has the sparsest jumps, in which reports 215 jumps with a jump frequency of 0.303. The rightmost column reports the mean statistic of the jumping for five financial indices. There are significantly more downside jumps (133) than upside jumps (115), implying a more likely downside jump risk in financial sectors. In addition, the whole sectors experience an average of 248 jumps, with a jump frequency of 0.350. Also, we report the statistics of jump risk measures in Panel B. To compare sizes, we focus on the absolute values of these measures. The results show that the securities sector has the largest average $|BJ_t|$, 12.793×10^{-3}; and the banking sector has the smallest average jump variation, 8.676×10^{-3}. Similarly, the securities sector performs the largest average $|PJ_t|$ and $|NJ_t|$, and the banking sector behaves the lowest $|PJ_t|$ and $|NJ_t|$, which implies that the jump risk of the securities sector is greater than the jump risk of the banking sector on average.

Furthermore, Fig. A.5 plots the pairwise jumps between these five sub-sectors. The diagonal subplots display the distributions of bilateral jumps for each player, and the remaining subplots show the pairwise jumps and OLS fitted lines between any two sub-sectors. The pairwise dots indicate that the jumps that occur in one index coincide with jumps in another index within the same day. We can find many pairwise dots falling in the first and third quadrants of the coordinate system, indicating that these jump behaviors are often synchronized. Also, the OLS fitted lines reflect positive correlations of jump risk between sectors.

Appendix B. Dynamic jump risk contagion topology analysis

Fig. B.6 presents the time series of the net directional connectedness ("Net contribution") which is the difference between the directional connectedness "To others" and "From others". From the net contagion of bilateral jump risk in different sub-sectors (black solid line), we can find large dynamic variabilities in the roles assumed by different sectors at any given time. At the beginning of the sample period, the net contributions of the real estate, securities, and diversified financial sectors are all less than zero (gray area) and play the net recipients of bilateral jump risk, while the net contributions of the insurance and banking sectors are both greater than zero (white area) and play net contributors of bilateral jump risk. Over the sample period, the five financial sub-sectors perform fluctuating net contribution paths with alternate roles across the risk network. At the end of the sample period, the real estate, securities, and banking sectors are the three net risk contributors, while the diversified finance and insurance sectors possess the opposite results.

More importantly, when we decompose the bilateral jumps into upside and downside jumps and subsequently consider their risk contagion net contributions separately, several asymmetric results emerge clearly in each sub-sector. For example, the real estate sub-sector has a longer positive period of net contribution of downside jump risk, playing a net contributor, from late 2019 to November 2021. However, the net contribution of its corresponding upside risk is almost always negative over the same period but experiences a sharp positive shock in October 2021. Overall, although both net contribution paths exhibit oscillatory characteristics, the net contagions of upside and downside jump risks in real estate show a clear divergence. As for the securities sub-sector, it has two positive net contribution periods to the downside jump risk contagion for the whole financial sector. One appears from the beginning of the sample period to November 2019, and the other

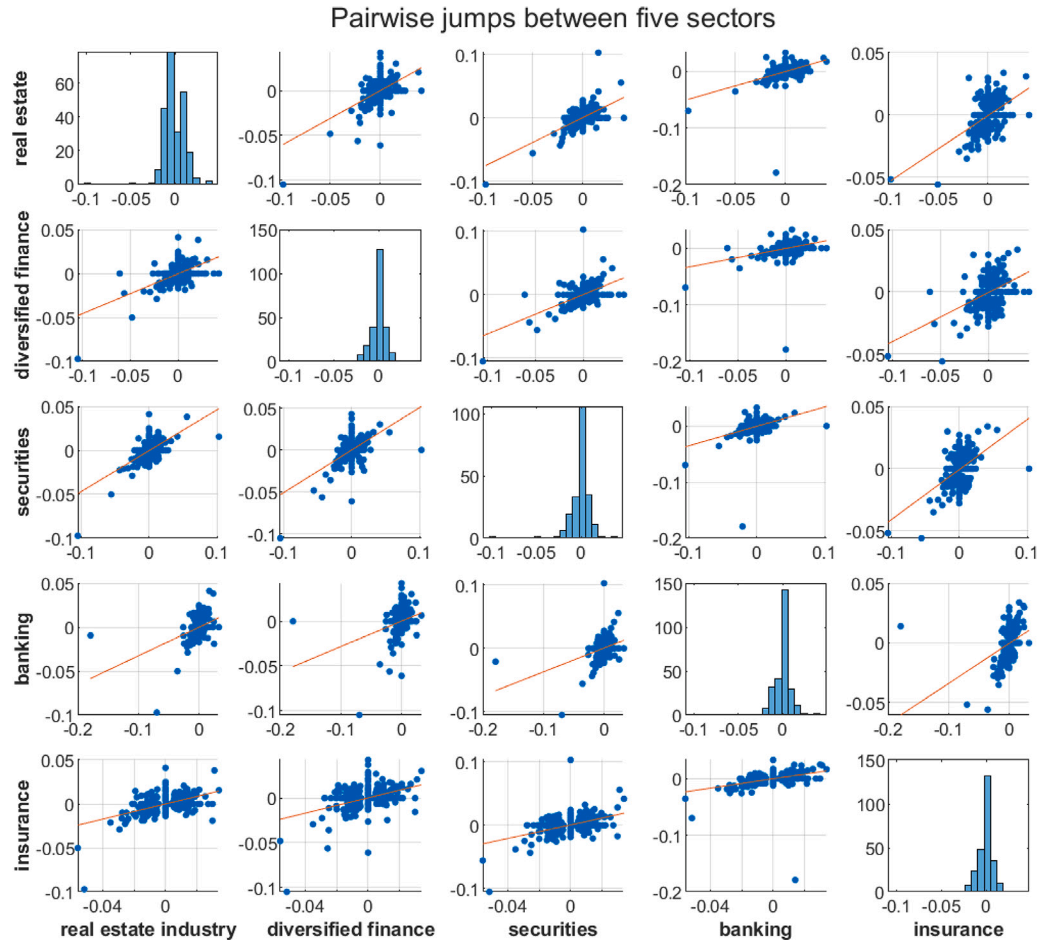

Fig. A.5. Pairwise jumps between five sectors.

takes place from April 2020 to January 2021. However, even the securities sector though has two positive net contribution periods for upside jump risk contagion, one of these two periods occurs from September 2019 to July 2020 and the other from August 2020 to March 2021, both of which are mismatched with the net contribution period of downside risk contagion. The same characteristics can be found in the remaining three sample sub-sectors.

In Fig. B.7, we display the rolling-sample dynamic jump risk contagion topology structures.[6] As can be seen from the subplots on the left column, the two thickest arc ribbons of purple and brown stand out in the early of the sample period, i.e. July 1, 2019, which is mainly caused by the relatively higher jump risk contagion between the banking and insurance sectors. Along with the outbreak of COVID-19, the middle-left subplot(02/28/2020) shows that the cross-sectoral arc ribbons possess similar coarseness and are intertwined with each other. This reveals that different sectors have similar scales of directional risk contagion. On September 30, 2019, the bilateral jump risk network shows several thickest blue, purple, and brown arcs that organize the real estate, insurance, and banking sector visibly together. This feature demonstrates the presence of tight bilateral jump risk contagion between the three sectors. In addition, the thickest red and green arc ribbons connect the securities sector to the diversified finance sector, implying a closely bilateral jump risk contagion between these two sectors.

Specifically, we construct dynamic network topologies separately for upside jump risk and downside jump risk. Several time-varying heterogeneity results emerge in the remaining subplots of Fig. B.6. We can find that in the upside jump risk network (upper-middle subplot) on July 1, 2019, the thickest purple and brown ribbons connect the insurance and banking sectors, while the thickest blue and red arcs connect the real estate and securities sectors. Next, the upside jump risk network (middle-middle subplot) during the COVID-19 pandemic indicates that the upside jump risk interaction between the real estate and insurance sectors is weakening. Meanwhile, the securities and diversified finance sectors have the highest level of upside risk contagion between them. The lower-middle upside jump risk network indicates that the risk interaction between the banking and real estate sectors is most prominent at the end of the sample period, followed by the interaction between the securities and the diversified finance sector. In addition, the upper-right subplot shows that the cross-sectional arc ribbons of the downside jump risk network intersect each other and possess about the same thickness, which indicates that the cross-sectional risk contagion between them is comparable. Also, the middle-right subplot(02/28/2020) shows similar characteristics to the upper-right, except that the ranking of the sources of downside jump risk from others in each financial sector has shifted. From the lower right jump risk network, we can find the thickest brown, blue, and purple ribbons linking the banking, real estate, and insurance sectors implying they have more interaction among them. More importantly, there are significant differences between the upside jump risk network and the downside jump risk network at any given time. Overall, these facts actually reveal the asymmetry between the upside jump risk and the downside jump risk.

[6] We consider the early (07/01/2019), middle (02/28/2020), and end (09/30/2021) of the sample period.

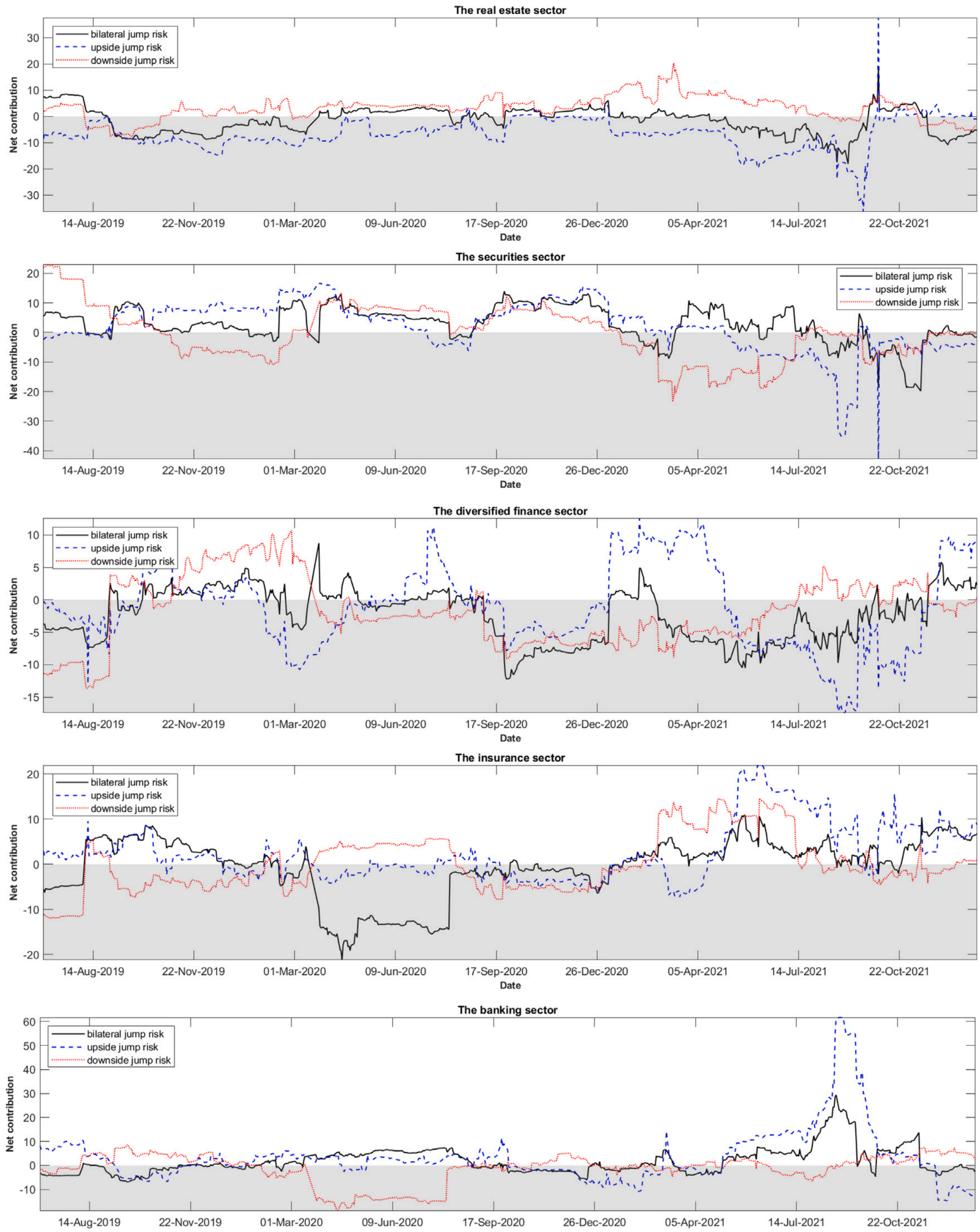

Fig. B.6. Dynamic net risk contribution for each sub-sector.

Appendix C. Variable explanation

Economic policy uncertainty (*EPU*)

Baker et al. (2016) develop an index of economic policy uncertainty (EPU) based on newspaper coverage frequency and find economic policy uncertainty has sizable effects on stock price volatility, investment rates, and employment growth. The existing literature on economic policy uncertainty focuses on the impact of the EPU on macro and micro levels, stock markets, corporate behavior, and risk management (Karnizova and Li, 2014; Bernal et al., 2016; Brogaard and Detzel, 2015; Bonaime et al., 2018). Since no scholars have so far explored the impact of EPU on jump risk contagion, this article aims to fill the gap. The EPU index for China can be downloaded from http://www.policyuncertainty.com.

The growth rate of money supply growth rate (*M2*)

We use the broad money supply, say M2, as a proxy variable for the central bank's money supply. M2 refers to cash circulating plus corporate deposits, residential savings, and other deposits, usually reflecting money supply and future inflation. We obtain the monthly growth rate of M2 from the WIND financial database.

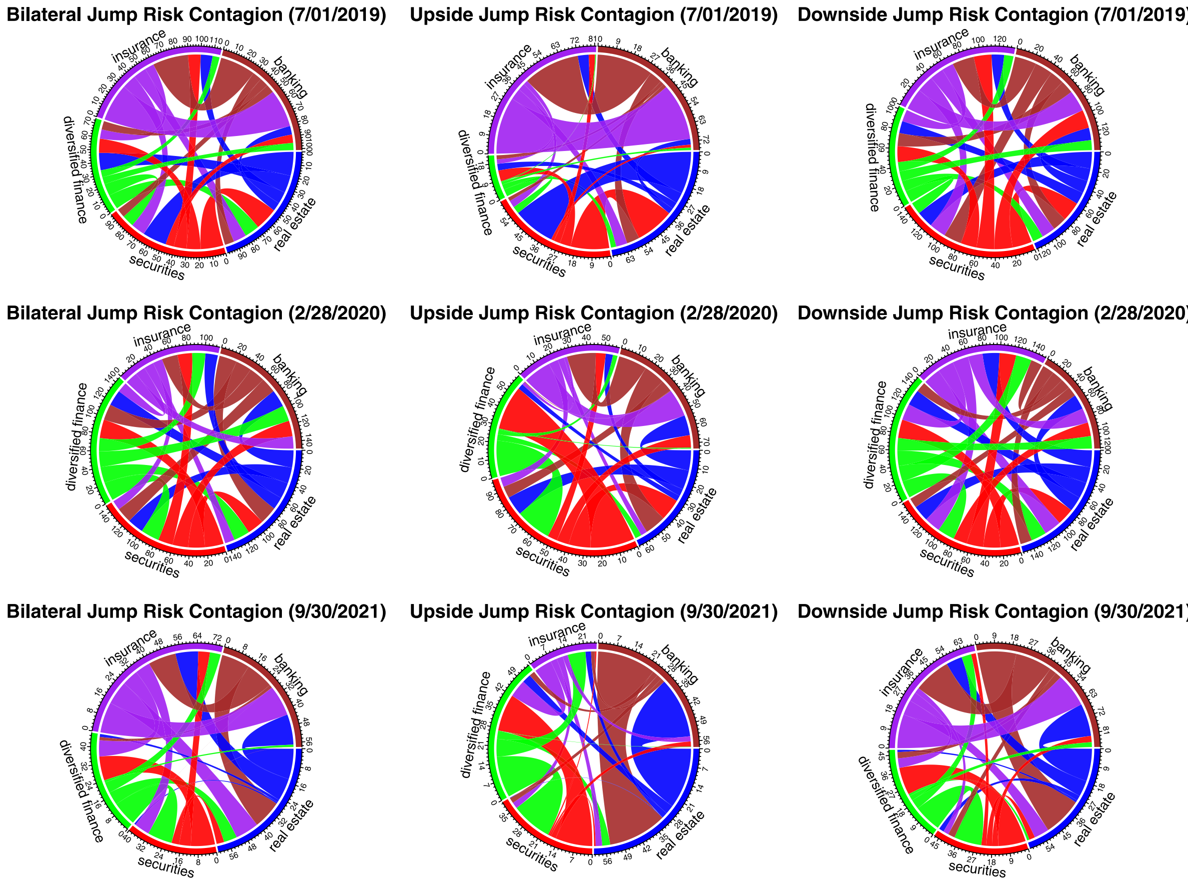

Fig. B.7. Rolling-sample dynamic jump risk contagion topology structures.

The growth rate of Aggregate Financing to the Real Economy (*AFRE*)

Aggregate Financing to the Real Economy, published by the Central Bank of the People's Republic of China, refers to the outstanding financing provided by the financial system to the real economy at the end of a period (monthly/quarterly/annual), which reveals the financing support to the real economy. The Chinese government employs it as a liquidity measurement tool in the economy to abet the formation of monetary policy. We obtain the growth rate of AFRE from the WIND financial database.

Financial sector valuation (*PB*)

Financial sector valuation reflects the fundamentals of the financial sector. We calculated the simple average of the five financial sector PB ratios as the PB variable in the regression. We can download the daily PB ratios for each sector from the CSMAR database.

Turnover (*TURNOVER*)

Turnover reflects investors' trading sentiment and liquidity for financial sector stocks. We calculated the simple average of five sub-sector trading turnover as the TURNOVER variable in the regression. We can obtain the daily market turnover data for each sector from the WIND financial database.

Chinese financial conditions (*CFCI*)

The China Financial Conditions Index ($CFCI$), published by the Yicai Research Institute, is a comprehensive index reflecting the cost and availability of financing for a wide range of economic entities in China. The index value is a standardized score (z-score), with values above zero representing a relatively tight financial environment, otherwise, a loose financial environment. This variable can be added in the regression analysis of this study to control for the effects of price-based monetary policy. The data can be downloaded from http://www.cbnri.org/index/fci/.

Fig. C.8 plots the trajectories of the six explanatory variables in the regression analysis. EPU exhibits a pattern of oscillations down over the sample period. Both M2 and AFRE experience a cycle of climbing and falling. PB slowly moves upward before declining slightly in late 2019, then climbs continuously to 3 before experiencing a decline to remain around 2. TURNOVER exhibits a volatile upward and downward cycle throughout the sample period. CFCI declines continuously from −0.4 at the beginning of the period to below −1.4, then climbs rapidly and then slowly declines to remain at −1.0.

Appendix D. Unit root test

We perform the ADF unit root tests on the variables. The null hypothesis is the existence of at least one unit root. The Panel A results in Table D.10 show that all level variables accept the null hypothesis at the 1% significance level. However, all first-order difference variables in Panel B reject the null hypothesis, implying that these variables are all I(1) series.

Appendix E. Johansen cointegration test

We perform the Johansen cointegration test for each of the three groups of variables. the explanatory variables in Panel A, B, and C of Table E.11 are the dynamic total connectedness of the bilateral, upside, and downside jump risks, Bi_Cidx, Up_Cidx, Dn_Cidx, respectively. The first column is the null hypotheses of the numbers of co-integrating equations; The third, fourth and last columns show the trace statistic, critical value and test probability, respectively. These alternative variables groups all accept the null hypothesis of the existence of at most three co-integrating equations at the 0.05 level.

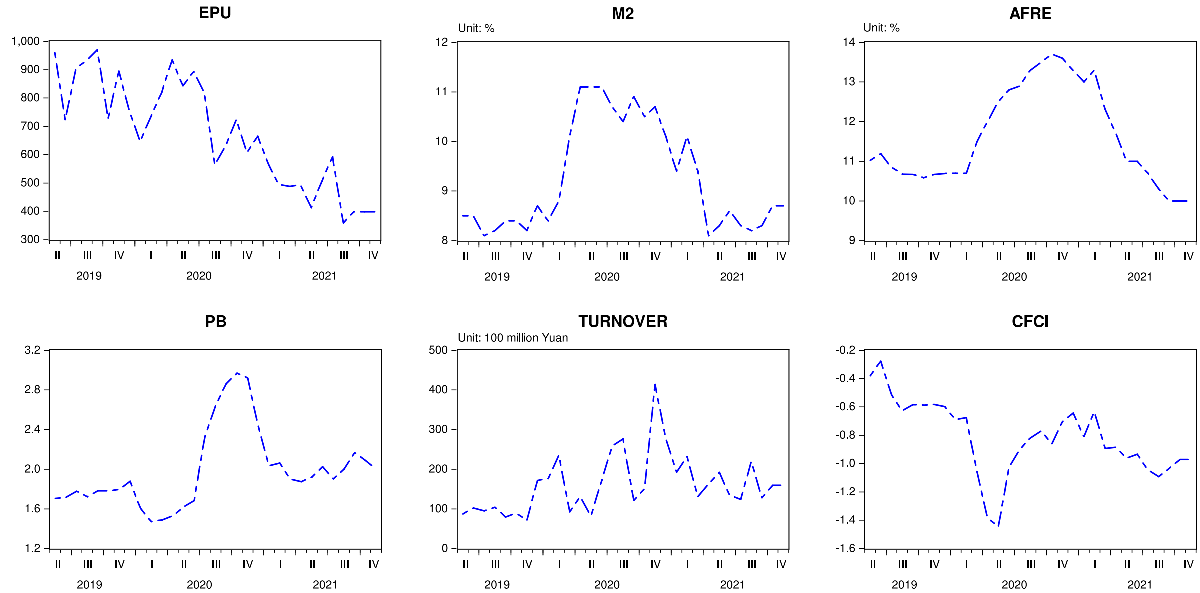

Fig. C.8. The movement path of variables.

Table D.10
Unit root test for variables.

Panel A: Level	Bi_Cidx	Up_Cidx	Dn_Cidx	EPU	$M2$	$AFRE$	PB	$TURNOVER$	$CFCI$
None	−0.470	−0.402	−0.823	−1.295	−0.086	−0.490	−0.331	−1.183	0.039
Intercept	−1.265	−1.451	−1.538	−1.857	−1.345	−3.548**	−2.407	−3.671**	−2.125
Intercept & Trend	−2.927	−2.859	−2.938	−4.205**	−1.281	−0.127	−2.430	−3.825**	−2.114
Panel B: 1st difference	Bi_Cidx	Up_Cidx	Dn_Cidx	EPU	$M2$	$AFRE$	PB	$TURNOVER$	$CFCI$
None	−4.715***	−4.155***	−6.161***	−7.996***	−4.601***	−3.180***	−3.241***	−6.434***	−4.771***
Intercept	−4.637***	−4.080***	−6.100***	−7.994***	−4.519***	−3.153**	−3.184**	−6.322***	−4.785***
Intercept & Trend	−4.669***	−4.280***	−5.989***	−7.903***	−4.538***	−3.322*	−3.146	−6.281***	−4.813***

Note: This table shows the ADF unit root test result for variables. The null hypothesis is the existence of at least one unit root. Panel A shows the results for the horizontal variables and Panel B shows the results for the first-order difference variables; * denotes rejection of the hypothesis at the 0.05 level.

Table E.11
Johansen cointegration test for variables.

Panel A:	$Bi\_Cidx, EPU, M2, AFRE, PB, TURNOVER, CFCI$			
H_0: # of CE(s)	Eigenvalue	Trace statistic	Critical value($\alpha = 0.05$)	Prob.
None *	0.909	194.655	125.615	0.000
At most 1 *	0.797	125.127	95.754	0.001
At most 2 *	0.726	78.819	69.819	0.008
At most 3	0.473	41.277	47.856	0.180
At most 4	0.382	22.685	29.797	0.262
At most 5	0.246	8.709	15.495	0.393
At most 6	0.018	0.534	3.841	0.465
Panel B:	$Up\_Cidx, EPU, M2, AFRE, PB, TURNOVER, CFCI$			
H_0: # of CE(s)	Eigenvalue	Trace statistic	Critical value($\alpha = 0.05$)	Prob.
None *	0.908	200.907	125.615	0.000
At most 1 *	0.801	131.739	95.754	0.000
At most 2 *	0.737	84.897	69.819	0.002
At most 3	0.575	46.201	47.856	0.071
At most 4	0.367	21.418	29.797	0.332
At most 5	0.235	8.17	15.495	0.447
At most 6	0.013	0.386	3.841	0.535
Panel C:	$Dn\_Cidx,\ EPU,\ M2,\ AFRE,\ PB,\ TURNOVER,\ CFCI$			
H_0: # of CE(s)	Eigenvalue	Trace statistic	Critical value($\alpha = 0.05$)	Prob.
None *	0.964	228.989	125.615	0.000
At most 1 *	0.838	132.57	95.754	0.000
At most 2 *	0.738	79.856	69.819	0.006
At most 3	0.528	41.003	47.856	0.189
At most 4	0.365	19.237	29.797	0.476
At most 5	0.156	6.071	15.495	0.687
At most 6	0.039	1.163	3.841	0.281

Note:Trace test indicates 3 cointegrating equations at the 0.05 level; * denotes rejection of the hypothesis at the 0.05 level.

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